

David Kreuter

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Senior Infrastructure Engineer | IaC, Observability & Self-Service Platform Builder

Infrastructure engineer with 10+ years of experience designing, building, and operating production infrastructure across cloud and on-prem environments. Expert in Infrastructure as Code using Terraform and Ansible, with deep Python development skills applied to custom orchestration middleware, self-service deployment platforms, and API-driven automation. Proven ability to define and standardize infrastructure patterns at scale--architecting a global provisioning framework managing 20,000+ virtual machines across three geo-redundant data centers. Passionate about observability, reliability, and empowering engineering teams to move fast with governance built in.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Architectural Scale:** Engineered a global server provisioning framework that deployed and managed 20,000+ virtual machines across cloud and on-prem environments; integrated Rubrik and VMware APIs to automate backup and compute placement, ensuring 100% RTO/RPO compliance at build time.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate the deployment of virtual machines to Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles, enabling teams to move fast with governance intact.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling--creating a self-service deployment platform for internal teams.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, enabling seamless deployment to Azure; designed the abstraction layer to be transparent to end-users, maintaining a consistent self-service interface while abstracting the underlying infrastructure provider.
- **Infrastructure Resilience:** Engineered and maintained a highly resilient infrastructure architecture spanning three identical, geo-redundant data centers to ensure maximum uptime, disaster recovery readiness, and application SLO compliance.
- **Observability & Operational Efficiency:** Developed an intelligent IP provisioning engine via InfoBlox API integrations with built-in monitoring and alerting, eliminating 1,000+ annual engineer hours and resolving a multi-year infrastructure bottleneck.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow delivering fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and configuration management, including optional, hardened SQL deployments at scale.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code--shaping infrastructure standards and influencing cross-functional technical decisions.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security policy updates into the server build cycle to ensure day-zero network isolation and a hardened security posture across the full stack.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in 'Read-Only' dry-run modes; implemented Least Privilege permissions and robust rollback protocols for stale object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/Memory topology to eliminate cross-node latency and restore system stability--applying deep distributed systems knowledge to root-cause analysis.
- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid cloud operations.
- **Security Architecture:** Engineered a secure reverse proxy solution to protect sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.

Network Engineer

- **Proactive Monitoring:** Developed service-aware alerting and observability frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection and structured response.
- **Network Administration:** Managed complex firewall rules and switch ACLs for a massive distributed network spanning 12 airports and three satellite offices, including DNS management and packet-level debugging.

- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture.
- **Enterprise Mobility:** Successfully onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geo-fencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics, driving observability across the stack.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Languages: Python, Go, Bash, PowerShell, SQL, YAML, Javascript, PHP

Automation/Orchestration: Terraform, CI/CD Pipelines, GitOps, Ansible Automation Platform, Middleware Development, Grafana, Gitlab, REST API, GraphQL, ServiceNow, Unit Testing

Infrastructure: Infrastructure as Code (IaC), Azure Cloud Services, VMware vSphere, OS Hardening (CIS/STIG), Nutanix AHV, Rubrik, DBaaS, NUMA Tuning, Check Point Firewalls, Active Directory

Networking: DNS, BGP, F5 Load Balancing, High Availability, VPN, VRRP, Intrusion Detection, Vulnerability Scanning, Meraki MDM

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

07/2025 - Present

Volunteer

Tiftt Nature Preserve

01/2025 - Present

Volunteer

716 Paws

01/2022 - Present

Thermal drone pilot for search and rescue missions.